

Juan David Muñoz-Bolaños

Rechenauerstraße 42, Innsbruck, Österreich (Tirol)

+43 676 9115145

juan.munoz@i-med.ac.at

SmarAct Metrology GmbH & Co. KG

z. Hd. Melissa Skrivanek

Oldenburg

Deutschland

Bewerbung als Applikationsingenieur (m/w/d), Laserinterferometrie und Vibrometrie

Sehr geehrte Frau Skrivanek,

hiermit bewerbe ich mich als Applikationsingenieur im Bereich Laserinterferometrie und Vibrometrie bei SmarAct Metrology. Ich bin ein ursprünglich aus Kolumbien stammender Physikingenieur und promoviere derzeit an der Medizinischen Universität Innsbruck. In meine künftige Rolle bringe ich fundierte Erfahrung in interferometrischer Messtechnik, Prototypenentwicklung und Datenauswertung ein.

Kern meiner Doktorarbeit ist ein interferometrisches Wellenfrontsensormesssystem, das ich eigenständig aufgebaut habe. Der optische Aufbau basiert auf einem Michelson-Interferometer (off-axis und phase-stepping mit Piezo-Aktuatoren); die Phasenrekonstruktion erfolgt über iterative Algorithmen (Zernike-Dekomposition), implementiert in Python und C++. Das System steuere ich in Echtzeit über LabVIEW/FPGA und nutze es täglich zur Aberrationsmessung. Zwei Publikationen (ACS Photonics, Biomedical Optics Express) dokumentieren die Ergebnisse. Parallel dazu habe ich ein Multiphotonen-Mikroskop komplett aufgebaut: Femtosekundenlaser-Integration, Galvo-Scanner-Ansteuerung, PMT-Signalerfassung und ZEMAX-Toleranzanalyse der Mikroskopobjektive.

Am Leibniz-IPHT in Jena habe ich zuvor einen 1064 nm Raman-Spektrometer-Prototyp von der optischen Auslegung über die Kalibrierung bis zur Publikation (Applied Sciences) durchgeführt. Dies entspricht einem vollständigen Prototypenzyklus, wie er auch bei SmarAct für den Übergang von Entwicklung zur Serie gefragt ist. Aus meiner früheren Tätigkeit als Machine Vision Engineer bringe ich zusätzlich Erfahrung in der Entwicklung optischer Inspektionssysteme und deren Überführung in die Serienproduktion mit.

Mein frühestmöglicher Eintrittstermin ist der 1. Juli 2026. Ich bin uneingeschränkt arbeitsberechtigt und freue mich auf ein persönliches Gespräch.

Mit freundlichen Grüßen,

Juan David Muñoz-Bolaños

Doktorand, Medizinische Universität Innsbruck

Juan David Muñoz-Bolaños

Optical System Engineer

Rechenauerstraße 42, Innsbruck, Österreich (Tirol)

+43 676 9115145

juan.munoz@i-med.ac.at | [LinkedIn Profile](#)



Summary

Physicist and optical engineer with deep expertise in **laser interferometry, wavefront sensing, and precision optical metrology**. Proven track record in end-to-end prototype development, from optical concept and simulation (ZEMAX, Python) through hardware-software integration (Python, C++, LabVIEW) to validated, publication-grade systems. Strong analytical skills and experience transitioning R&D concepts into robust, reproducible measurement setups. Author of three peer-reviewed publications in applied optics and photonics.

Professional Experience

Jan 2022 to Present **Medical University of Innsbruck** Innsbruck, Austria
PhD Researcher, Laser Interferometry & Adaptive Optics

Interferometric Wavefront Sensing: Designed and built a Michelson-type interferometric wavefront sensor (off-axis and phase-stepping with piezo actuators) for adaptive optics microscopy; implemented iterative phase retrieval algorithms and a C++/Python/LabVIEW control stack for scientific cameras and deformable mirrors.

Prototype Development & Validation: Led the end-to-end assembly and commissioning of a custom nonlinear (multiphoton) microscope, including fs-laser integration, galvo-scanner control, and PMT signal acquisition, achieving diffraction-limited resolution.

Physical Modelling & Simulation: Performed systematic ZEMAX tolerance analysis and wave-field simulations (Zernike decomposition) to characterize aberration sensitivity of custom microscope objectives.

Technical Documentation: Published three peer-reviewed articles in ACS Photonics, Biomedical Optics Express, and Applied Sciences, demonstrating whitepaper-level scientific writing.

Jun 2020 to Dec 2021 **Leibniz Institute of Photonic Technology (IPHT)** Jena, Germany
R&D Research Assistant, Photonics & Spectrometer Prototyping

Spectrometer Prototyping: Designed, built, and calibrated a dispersive 1064 nm fiber-probe Raman spectrometer for tissue characterization; published results in Applied Sciences.

Data Acquisition & Analysis: Developed 'SpectraMap' (custom Python package) for supervised ML classification of hyperspectral Raman images, optimizing measurement throughput and evaluation.

Mar 2018 to Mar 2019 **INTECOL S.A.S.** Medellín, Colombia
Junior Machine Vision System Engineer

Optical System Design: Engineered custom illumination (dark-field, bright-field, coaxial) and Halcon pipelines for bottle labeling lines (liquid levels, cap sealing); implemented Cognex VisionPro for automatic part recognition in automotive lines.

R&D-to-Production: Transitioned prototype vision setups to production-ready deployments; performed Gage R&R assessments and feasibility studies for client sign-off.

Technical Competencies

Laser Interferometry & Metrology: Wavefront sensing, phase retrieval, Michelson interferometry, adaptive optics, precision alignment of laser-based measurement systems.

Prototyping & Optical Design: End-to-end system build from concept to validated prototype. ZEMAX/OpticStudio modelling, tolerance analysis, opto-mechanical integration, component qualification (Thorlabs, Edmund Optics, Hamamatsu).

Data Acquisition & Simulation: Python, C++, LabVIEW/FPGA, MATLAB for real-time data capture, signal processing, physical modelling, and instrument control. Experience with JAX (GPU computing).

Technical Documentation: Three peer-reviewed publications (ACS Photonics, Biomedical Optics Express, Applied Sciences). Experienced in writing application notes, technical specifications, and whitepaper-style documents.

Industrial Optics: Halcon & Cognex VisionPro pipelines, custom illumination design, camera integration, Gage R&R, system commissioning on production lines.

Education

Jan 2022 to Present	Medical University of Innsbruck <i>Ph.D. in Adaptive Optics & Microscopy</i>	Austria
2019 to 2021	Friedrich Schiller University Jena <i>M.Sc. in Photonics</i>	Germany
2012 to 2018	University of Cauca <i>B.Sc. in Physics Engineering</i>	Colombia

Publications & awards

Awards: ASML & ZEISS Summer Schools participant; COLFUTURO Grant 2026 awardee.

Muñoz-Bolaños, J. D. et al. Confocal Raman Microscopy with Adaptive Optics. ACS Photonics 12(1), 176–184 (2025).

Muñoz-Bolaños, J. D., et al. Design of a dispersive 1064 nm fiber probe Raman imaging spectrometer. Applied Sciences, 14(11), 4726 (2024).

Sohmen, M., Muñoz-Bolaños, J. D., et al. Optofluidic adaptive optics in multi-photon microscopy. Biomedical Optics Express 14(4), 1562–1578 (2023).

Languages

English (Professional) · German (B2, intermediate/ working proficiency and ongoing) · Spanish (Native)

Soft skills

Collaboration & Communication: Communicated complex technical results through peer-reviewed publications, conference presentations (SPIE Photonics West), and international seminars within cross-functional R&D teams.

Problem Solving: Tackled complex, cross-disciplinary challenges resulting in three peer-reviewed publications and a conference proceeding presented at SPIE Photonics West, San Francisco.

Adaptability: Rapidly integrated into three distinct research environments across three countries (University of Cauca, Leibniz-IPHT Jena, Medical University of Innsbruck), each with different workflows, toolchains, and scientific cultures.

Hobbies

Mountain Biking, Dancing Bachata, Learning Harmonica, Learning German.

References

Prof. Monika Ritsch-Marte

Head of Institute of Biomedical Physics
monika.ritsch-marte@i-med.ac.at

Medical University of Innsbruck

Dr. Christoph Krafft

Spectroscopy Group Leader
christoph.krafft@leibniz-ipht.de

Leibniz Institute of Photonic Technology

AGREEMENT FOR GUESTS

to carry out a practical training
with

Mr. **Juan David Munoz Bolanos** Date of birth: **1994-06-04**
temporarily living at **Emil-Wölk-Str. 7, 07747 Jena** (guest),

Student of the Friedrich-Schiller-University Jena, Abbe School of Photonics, gets in execution of his practical course in the research department Spectroscopy and Imaging, working group Raman and Infrared Histopathology (0104) from 2020-06-02 up to 2020-12-31 admission to the necessary rooms within Leibniz-IPHT.

The use of further devices, media and other infrastructure takes place in agreement with the Heads of the respective departments.

Mr. Juan David Munoz Bolanos has to observe the security regulations for working and other regulations valid in Institute as well as instructions of the laboratory and clean room responsible persons of the Leibniz-IPHT.

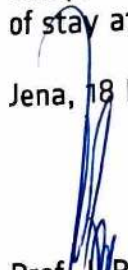
Mr. Juan David Munoz Bolanos has to treat all affairs and procedures strictly confidential. Copies, duplicates and others of business and technical information of each kind are forbidden, unless these are sanctioned by the responsible Department Heads.

Publications, reports and lectures about topics which belong together with the fields of work of Leibniz-IPHT require the previous written agreement of the Executive Board.

The liability of Mr. Juan David Munoz Bolanos is limited to cases of wilful action and gross negligence.

Mr. Juan David Munoz Bolaos is insured against accidents on the Friedrich-Schiller-University Jena, Abbe School of Photonics in the context of his/her practical course during his/her length of stay at Leibniz-IPHT.

Jena, 18 May 2020


Prof. Dr. Popp
Chairman
of Leibniz-IPHT


F. Sonderrmann
Executive Member
of Leibniz-IPHT


Juan David Munoz Bolanos
Guest

OFFICIAL TRANSLATION OF DOCUMENTS WRITTEN IN SPANISH, MADE 20
DECEMBER 2018 BY JUAN CARLOS OJEDA-GARCÉS, OFFICIAL TRANSLATOR
BY RESOLUTION No. 1171 ISSUED BY THE MINISTRY OF JUSTICE, REPUBLIC
OF COLOMBIA.



UNIVERSIDAD DEL CAUCA
ON BEHALF OF THE
REPUBLIC OF COLOMBIA
AN BY AUTHORIZATION OF THE MINISTRY OF NATIONAL EDUCATION
CONSIDERING THAT

JUAN DAVID MUÑOZ BOLAÑOS
CC 1.061.772.278 OF POPAYAN

HAS COMPLIED WITH ALL THE STATUTORY AND LEGAL REQUIREMENTS,
GRANTS THE TITLE OF

PHYSICAL ENGINEER

WITH ALL THE RIGHTS, PRIVILEGES AND DIGNITIES THAT AUTHORIZE HIM
FOR THE PROFESSIONAL EXERCISE

POPAYAN, 16 MARCH 2018

REGISTERED IN THE DIPLOMA BOOK No. 082 FOLIO 437 DIPLOMA No. 396
RESOLUTION No. R-266-07 MINUTE No. 14-07

(SIGNED)

THE PRINCIPAL OF THE UNIVERSITY
THE DEAN OF THE FACULTY
THE SECRETARY GENERAL OF THE UNIVERSITY

Juan Carlos Ojeda - Garcés
TRADUCTOR INTERPRETE OFICIAL
RESOLUCIÓN No 1171
MINISTERIO DE JUSTICIA 1997



URKUNDE

Die Friedrich-Schiller-Universität Jena verleiht durch die
Physikalisch-Astronomische Fakultät
mit dieser Urkunde

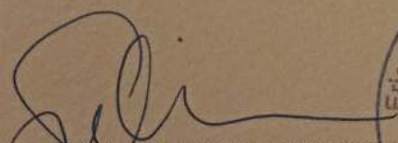
Herrn Juan David Munoz Bolanos
geb. 04.06.1994 in La Cruz (Kolumbien)
den Hochschulgrad

MASTER OF SCIENCE (M.Sc.)

Er hat die Masterprüfung (120 ECTS)
im Studiengang »Photonics«

bestanden.

Jena, 20.12.2021


Prof. Dr. Christian Spielmann
Dekan




Prof. Dr. Silvana Botti
Vorsitzende des
Prüfungsausschusses

Zertifikat

Muñoz-Bolaños JUAN DAVID

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Schriftliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 16.09.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ö s d

österreich schweiz deutschland

ID-Nummer: ZB2Ö25s34291

Zertifikat

Juan David MUÑOZ BALAÑOS

geboren am 04.06.1994

in La Cruz / Kolumbien

hat das Modul **Mündliche Prüfung**

ÖSD Zertifikat B2

am Prüfungszentrum

BFI Tirol Bildungs GmbH

in Innsbruck / Österreich

bestanden



Datum: 11.08.2025

für das ÖSD: Prüfungsvorsitzende/r
Maria VALLAZZA



ÖSD ist Mitglied von



ösd

österreich schweiz deutschland



ID-Nummer: ZB2Ö25m40434

Juan Munoz Bolanos

has participated in the

**ZEISS European
Summer School**

Lithography Optics &
Semiconductor Technologies

26 - 27 June 2025

ZEISS Semiconductor Manufacturing Technology, Oberkochen, Germany

Thomas Marschner

Dr. Thomas Marschner
Program Manager Funding Projects

Thomas Stammer

Dr. Thomas Stammer
CTO & Head of Product Strategy



INTECOL

OPTIMIZING THE RHYTHM OF THE WORLD

Medellin, February 4, 2019

TO WHOM IT MAY CONCERN

Integración Tecnológica Industrial Intecol S.A.S. with Tax ID (NIT) 811.041.410-4 certifies that Mr. JUAN DAVID MUÑOZ BOLAÑOS, identified with Citizenship ID No. 1.061.772.278, has been employed by this company since July 3, 2018, under an indefinite term contract, serving in the position of Junior Project Engineer;

earning a salary of One million five hundred thousand pesos (\$1,500,000) plus three hundred thousand pesos (\$300,000) mobility allowance.

I will gladly address any additional information at the phone number 316 3322.

We appreciate your attention,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Administrative and Financial Director

Integración Tecnológica Industrial Intecol S.A.S.
www.intecol.com.co
Medellin Calle 42 #63c-82
PBX. (574) 316 3322 Cell: (57) 301 430 2532

INTECOL

DEN RHYTHMUS DER WELT OPTIMIEREN

Medellín, den 4. Februar 2019

AN ALLE, DIE ES BETRIFFT

Integración Tecnológica Industrial Intecol S.A.S. mit der Steuernummer (NIT) 811.041.410-4 bescheinigt, dass Herr JUAN DAVID MUÑOZ BOLAÑOS, identifiziert mit dem Personalausweis Nr. 1.061.772.278, seit dem 3. Juli 2018 in diesem Unternehmen beschäftigt ist. Er hat einen unbefristeten Arbeitsvertrag und ist in der Position als Junior-Projektingenieur tätig;

er bezieht ein Gehalt von einer Million fünfhunderttausend Pesos (\$1.500.000) zuzüglich dreihunderttausend Pesos (\$300.000) Mobilitätszulage.

Für weitere Informationen stehe ich Ihnen gerne unter der Telefonnummer 316 3322 zur Verfügung.

Wir danken für Ihre Aufmerksamkeit,

INTECOL

NIT. 811.041.410-4 - Tel.: 3163822

LENY ADRIANA RUIZ PINO

Verwaltungs- und Finanzdirektorin

Integración Tecnológica Industrial Intecol S.A.S.
www.intecol.com.co
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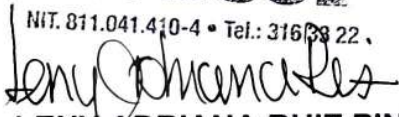
Medellín, 4 de Febrero de 2019

A QUIEN PUEDA INTERESAR

Integración Tecnológica Industrial Intecol S.A.S. con NIT.811.041.410-4 certifica que el señor **JUAN DAVID MUÑOZ BOLAÑOS**, identificado con cédula de ciudadanía N°1.061.772.278, labora en esta compañía desde el 3 de Julio de 2018, con un contrato a término indefinido, desempeñando el cargo de Ingeniero de proyectos Junior; devengando un salario Un millón quinientos mil pesos ml (\$1.500.000) más trescientos mil pesos ml (\$300.000) de movilidad.

Cualquier información adicional con gusto la atenderé en el teléfono 316 3322.

Agradecemos su atención,

INTECOL
NIT. 811.041.410-4 • Tel.: 316 3322,

LENY ADRIANA RUIZ PINO
Directora Administrativa y Financiera
Integración Tecnológica Industrial Intecol S.A.S.

Passports № / Passport No

MUNOZ BOLANOS

COLOMBIANA

Sexo / Sex
Lugar de nacimiento / Place of birth

M LA CRUZ COL

Fecha de expedición / Date of issue

11 DIC/DEC 2018

Fecha de vencimiento / Date of expiry

10 DIC/DEC 2028

Autocidad / Authority

G. ANTIOQUIA

Firma del titolare / Titolare's signature

Chas. F. Smith

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